Can Minkowski tensors of a porous microstructure characterize its permeability?

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Abstract

We show that the permeability of porous media can be reliably predicted from the Minkowski tensors (MTs) describing the geometry. To this end, we consider a large number of simulations of flow through periodic unit cells containing complex shaped obstacles. The prediction is achieved by training a deep neural network (DNN) using the simulation data with the MT elements as attributes. The obtained predictions allow for the conclusion that MTs of the pore microstructure contain sufficient information to determine the permeability, although the functional relation between the MTs and the permeability could be complex.

Key words: Porous media, machine learning, shape description, Minkowsi tensors, deep neural network

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1. Introduction

The characterisation of the microstructure of a porous medium is an unsolved problem for nearly a century. While the influence of the porosity on the permeability is well studied [1, 2, 3], the effect of the shape of the porous microstructure itself on permeability has remained unclear. The problem is two-fold: (1) shape, in general, is impossible to unambiguously describe using a practically finite number of parameters; (2) it is unclear how to identify shape descriptors that are relevant to the permeability of the porous microstructure. In this paper we identify Minkowski Tensors (MTs) [4] as shape descriptors for the porous microstructure by showing that MTs are sufficient to determine its permeability. To demonstrate this, we use the data generated from a large number of computational fluid dynamics (CFD) simulations of flows through 2D periodic porous media (with complex shaped obstacles in each periodic cell).

As shape descriptors, one could, for example, use spatial discretizations (like voxel data from tomograms or the points on a reconstructed surface) or parametric representations (such as spherical harmonics). While the first class of descriptors depends on the spatial resolution, and therefore may require processing a large amount of data, the latter is not applicable to general porous structures. Simple representations of shape, relevant to flow through porous media, such as the widely used 'tortuosity' [5] or Euler characteristics [6], while being very important and relevant to porous media, are insufficient to fully characterize permeability. Also, experimentally obtained

fitting parameters are required to related permeability to these characterizations [7].

Minkowski Tensors have been introduced [4] as a shape descriptor for a variety of different complex shaped media such as cellular structures and granular media. In granular media, for example, MTs reveal properties of packings that have been difficult to obtain by any other descriptors [8]. While the idea to use Minkowski functionals (MTs are a generalization of the functionals [9]) to characterize porous permeability [10], by now there is no evidence showing that MTs can reliably characterize the permeability [6, 11].

In this study, we employ a Deep Neural Network (DNN) in order to capture the relation between MTs and permeability without having to provide a functional form a' priori. Previous studies (for example [11]) have shown that linear relation between Minkowski functionals and the porous permeability may not exist. Deep Neural Networks are known to capture complex nonlinear relations and its use to relate MTs to permeability is practical. We use 2D, periodic domains, each containing a complex shaped obstacle, generated by a random process. Subsequently, we simulate the flow through each of these representative elementary volumes (REVs). In total, we perform CFD simulations for 10^5 unique shapes of invariant volume (to represent constant porosity across our samples) and for different flow velocities ($Re < 1$). From the simulaed flow data, we determine the porous permeability for each of the REVs. Additionally, we compute the MTs for each of the shapes in the REVs. Finally, we train a DNN [12] model to predict the permeability as a

function of the MTs' Eigen values and vectors using a portion of the data and validate the model against the remaining data.

The obtained model can predict permeability to high accuracy establishing that the MTs for a given structure are sufficient to characterize the permeability for at least the considered class of porous media. The fact that MTs can be used to represent arbitrary shapes, thus gives us the confidence to hypothesize that MTs can, in principle, determine permeability in general, simply connected porous media. In this paper we present the CFD simulation setup used for generating data, the description of the MTs used as attributes for the DNN training and the set up of the DNN used to train the model for the porous permeability. Further, we present the results for shapes of different complexity, discuss the individual prediction capacity of each of the MTs exclusively and present the training accuracy for increasing data size. The data and the DNN parameters are provided as supplementary material.

2. Shape generation and flow simulation

In order to test if the MTs can sufficiently predict porous permeability, we have to consider a system which is complex enough, but computationally not too challenging, so that a large data set can be computed in a reasonable period of time. Also the system we use should account for shape effects that are beyond the obvious dependence of permeability on the porosity [1, 13]. Considering these conditions we generate square 2D domains with periodic

boundary conditions containing a complex simply connected arbitrary shaped obstacle of constant area. The obstacle is assigned no slip wall boundary condition. Each such domain is therefore assumed to form the REV for a porous medium.

Figure 1 illustrates the generation of shapes. Ten points are spawned at equal angular intervals with randomly generated radial coordinates. A Fourier parametric curve is fitted to these points using least square minimization. Similar approaches are widely used in reconstructing arbitrary non-convex shapes [14, 15]. The parametric representation of the shape of the obstacle is of the form:

$$x(\theta) = \frac{1}{2} X_0 \Sigma_{i=0}^{m} X_i^c \cos(i\theta) + X_i^s \sin(i\theta) \tag{1}$$

$$y(\theta) = \frac{1}{2} Y_0 \Sigma_{i=0}^{m} Y_i^c \cos(i\theta) + Y_i^s \sin(i\theta). \tag{2}$$

For $\theta \in [0, 2\pi)$, x and y describe a shape of order m. The coefficients $X_0/2$ and $Y_0/2$ are the coordinates of the centroid of a shape. The volume of a given shape can be computed by:

$$A = \int_0^{2\pi} y(\theta) x'(\theta) d\theta, \tag{3}$$

where $x' = dx/d\theta$. The shapes are scaled such that the area is maintained to an invariant value of 0.5 for all shapes (to keep the porosity fixed, $\phi = 0.5$)

and the centroid $(X_0/2, Y_0/2)$ is translated to the center of the unit square domain. Shapes with regions outside the square, occurance of sharp spikes and self intersections are eliminated during the data generation. Different values for m, namely 1, 2 and 3, are considered, and for each of these orders (m), 100,000 shapes are generated. For $m = 1$ only ellipses are obtained, while higher values of m results in more complex shapes. Figure 1(b) shows shapes of different order generated with the same set of generator points (positioned at equal θ intervals but at random radial locations).

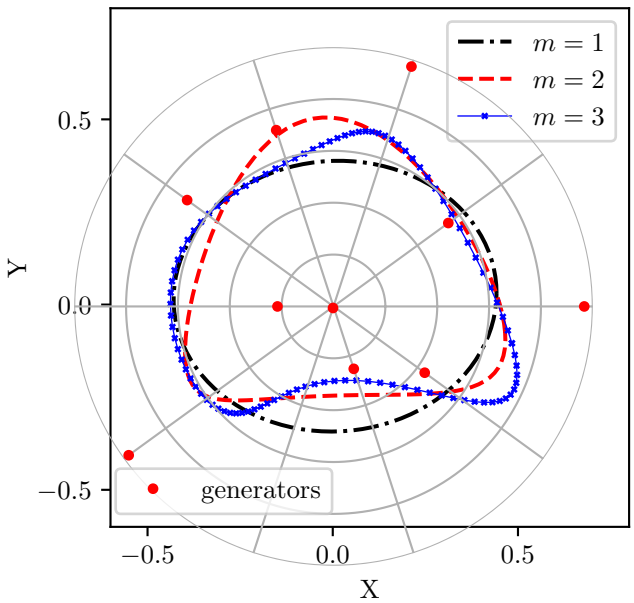


Figure 1: Shapes generated by parametric curves according to Eq. 2 to points located at random radial locations

The flow through each of the REVs is then simulated with the Gerris flow solver [16] employing the finite volume method (FVM). This solver was chosen because it uses an adaptive quadtree mesh which eliminates the need

for additional meshing and pre-processing difficulties. For each REV, three different pressure gradient values were applied to drive the flow in the horizontal direction. The largest pressure gradient magnitude was chosen such that the corresponding Reynolds number is 1 ($Re = \rho U r/\mu$, where r is radius of the circle of equal area as the shape, U is the superficial velocity and ρ and μ are the density and viscosity of the fluid). Therefore, we, expect the flow regime to not deviate substantially from the Darcy regime at least at the smallest pressure gradient case. For each REV, Darcy's equation for porosity reads:

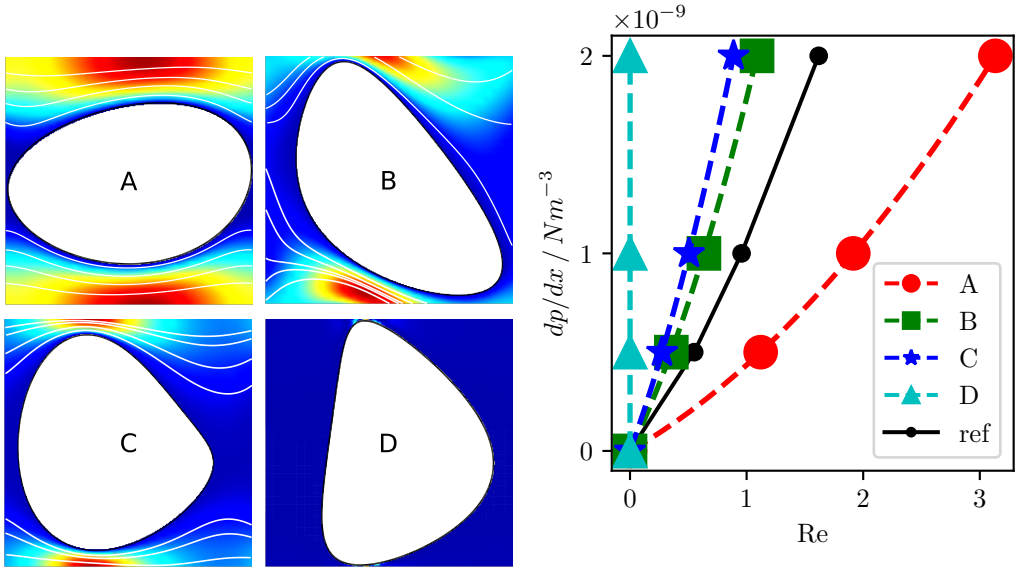
$$\frac{dp}{dx} = -\frac{\mu}{K}U. \tag{4}$$

Here dp/dx is the applied pressure gradient in the x-direction, U is the obtained superficial velocity in x-direction, μ is the viscosity, and K is the permeability (assuming an isotropic porous medium). For each REV, we fit the following 2nd order polynomial for the pressure gradient, dp/dx, as a function of U:

$$\frac{dp}{dx} = -\frac{\mu}{K}U - \frac{\rho}{K_1}U^2, \tag{5}$$

where ρ is the density of the fluid and K_1 is the inertial permeability, also known as the Forchheimer permeability term [17]. The slope of this curve at the origin ($U = 0$), gives the coefficient of the linear term in U, namely μ/K. For a given viscosity, μ, the value of K can be thus computed for each REV. The coefficients of the second order term, obtained by fitting the simulation

data, are provided in the supplementary material.



(a) (b)

Figure 2: Flow through four exemplary shapes for $m = 2$: (A) and (D) correspond to the shapes with maximum and minimum permeability in the dataset, respectively. The contours in (a) correspond to the velocity component in x-direction and the white lines represent the flow streamlines. The plot (b) shows the fit of Eq. 5 for the velocity values (normalised as Re) for varying pressure gradient (in x-direction). The solid line denotes the circular shape used as a reference shape.

Figure 2a shows four example shapes from the dataset including the extreme cases, i.e., the most permeable REV (A) as well as the least permeable REV (D) among the shapes with $m = 2$. The plot in Fig. 2b shows the Reynolds number (based on superficial velocity) against the pressure gradient for each of these example shapes and the polynomial fit to this data. For each of the flow simulations the permeability were thus obtained.

Figure 3 shows the distribution of the permeability values for each set of shapes corresponding to certain values of m. As m increases, we observe that

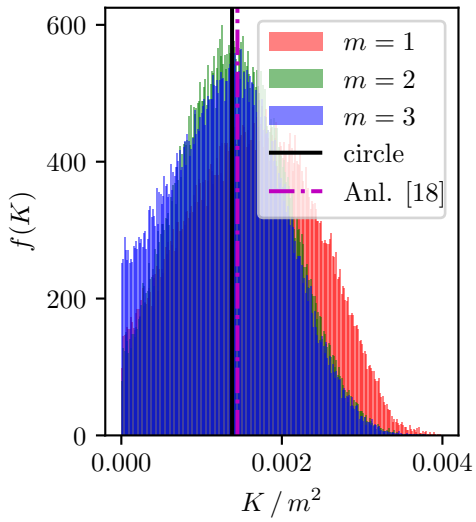


Figure 3: The probability density (f) of permeability values (K) for the shapes of different complexity (m) with constant porosity of 0.5. For each complexity class, 10^5 shapes were used. The solid line indicates the permeability for a circular geometry from simulation and the dashed line indicates the permeability for a circular shape determined analytically [18]

an increasing number of shapes in the dataset has zero permeability. This is because the probability of a shape blocking the flow increases as the number of free parameters determining the shape increases.

The permeability values of a similar periodic flow with a circular obstacle ($\phi = 0$) is marked with a solid line in Fig. 3 for reference. Also an analytical result [18] for the permeability of a periodic arrangement of circles ($\phi = 0.5$) in a square array is marked with a dashed line in the figure. The permeability of a circular obstacle is close to the value corresponding to the peak probability density of our dataset.

3. Minkowski Tensors

Minkowski tensors are generalizations of the Minkowski functionals [4, 10] and can characterize a shape using translation covariant and translation invariant tensors. Four linearly independent Minkowski tensors can be defined for two spatial dimensions and six for three spatial dimensions. Our study includes only two dimensional analysis. The MTs used for representing 2D shapes are as follows:

$$W_0^{2,0} = \int_{\Omega} \mathbf{r} \otimes \mathbf{r} \, dA \tag{6}$$

$$W_1^{2,0} = \frac{1}{2} \int_{\partial\Omega} \mathbf{r} \otimes \mathbf{r} \, dr \tag{7}$$

$$W_2^{2,0} = \frac{1}{2} \int_{\partial\Omega} \kappa(\mathbf{r}) \mathbf{r} \otimes \mathbf{r} \, dr \tag{8}$$

$$W_1^{0,2} = \frac{1}{2} \int_{\partial\Omega} \mathbf{n} \otimes \mathbf{n} \, dr. \tag{9}$$

Here, Ω represents the area chosen for integration (the obstacle) and $\partial\Omega$ represents the boundary. The vectors $\mathbf{r}$ and $\mathbf{n}$ are the position and the unit normal vector at the surface, respectively. Curvature at a point on the surface is denoted by κ. The MTs can be computed numerically using the open source software *Papaya* [9]. In this work, we employ the latter to compute the MTs.

Each MT is a symmetric, two-dimensional, 2nd order tensor and, thus, contains 3 independent scalar values. For the purpose of building the DNN

model we could directly use these elements as the features for the training. However, we use the Eigen values as well as the elements of the Eigen vectors as training features. This results in six features (scalar values) for each MT, making a total of 24 features. In the following section, we discuss the set up of the DNN and its training process.

4. The DNN model

We have chosen the *Keras* [19] library with a *TensorFlow* backend for our purpose. As mentioned in the previous section, we consider the Eigen values and Eigen vector elements of the 4 Minkowski tensors forming a 24 dimensional feature space for the DNN. The features are provided as input nodes to a DNN, followed by 7 hidden layers and an output layer with a single node (permeability values). We have chosen the *tanh* function as the activation function. Further details of the neural network architecture used in our study is provided in table 1. Additionally the training data and script is provided as an iPython notebook is made available as supplementary material.The L_2 norm of the difference between the predicted K vlaues and the actual K value for the trainign data is used as the *loss function* for the training process. The training of the DNN involves improving the parameters (weights and biases) of each of the nodes (neurons) of the network in order to minimize the loss function through several iterations. Each such iteration is termed an *epoch* in the machine learning parlance.

The available dataset is split into a *training set* and a *validation set*.

Parameter	Value
activation function	tanh
hidden layer type	dense
number of layers	9
hidden layer sizes	{24, 24, 48, 72, 32, 4, 1}
loss function	mean squared error
optimizer	stochastic gradient descent
initial learning rate	0.005
feature normalisation	$\min - \max$

Table 1: Training parameters used in training the DNN. For details of implementation, refer to the scripts in the supplementary material.

While the training set is used to minimize the loss function, the validation set is used to establish the accuracy of the prediction and to check for overfitting. We check if the error in the validation set and that in the training set diminishes at the same rate so that the model improves in accuracy. A decrease in error for the training set at a higher rate than the error for the validation set would mean that the model overfits the training data and, therefore, would be invalid on any data outside the training set. For a successful model, the right choice of training parameters is necessary to achieve a balance between bias and overfitting. Often it is required to experiment with the training parameters until a reasonable error convergence without overfitting is obtained.

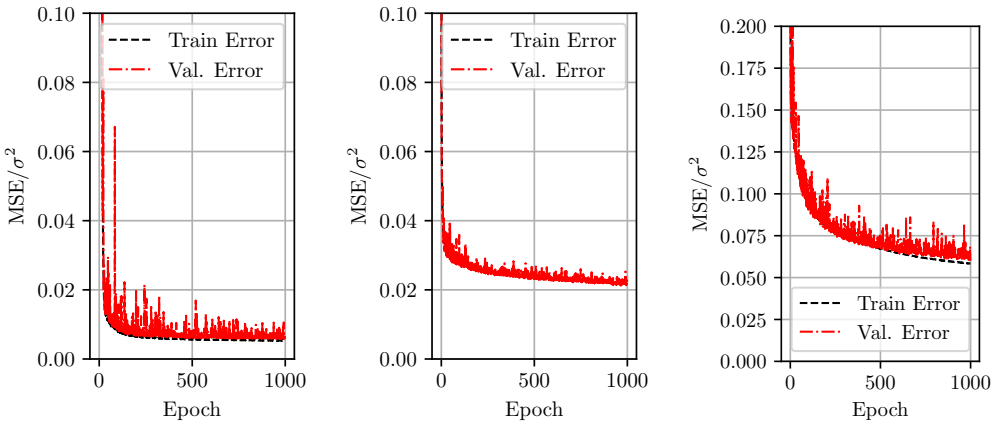
We show, in Fig. 4, that the model we trained can predict K within 6 % relative mean squared error (both training and validation dataset) relative to the total variance in the permeability values (see Fig. 3) for each of the

set of shapes. It may be possible to improve the parameters of the DNN and achieve better prediction of K using the MTs; with the achieved accuracy levels we assert that MTs are sufficient to relate the effects of shape (beyond porosity) to the permeability.

5. Results and discussion

Beyond the obvious dependency of permeability on porosity [1], a comprehensive dependence of permeability on shape has been a subject of intense research [20, 21, 22, 23, 24]. These studies, in addition to volumes of research undertaken in this area, reveal that the functional relationship between a shape descriptor and the permeability may not be mathematically simple to determine. Characterizing porous media using Minkowski functionals and tensors have been attempted in the past with simple fit functions and could not conclude that permeability is dependent on MTs [10, 11]. In other applications using MTs such as in structure of sphere packings, the higher order MTs have not revealed additional insights about the structures than those by the lower order MTs [8], raising the question whether these higher order measures have practical significance. In Fig. 4, we show that for shapes of different complexity (m), the same training parameters is unable to train the DNN models to the same accuracy. However, the overall accuracy is still high enough for practical purposes. We, however, have maintained the same DNN parameters for the sake of uniformity accross datasets.

Minkowski tensors provide much more information about the shape than



(a) $m = 1$ (b) $m = 2$ (c) $m = 3$

Figure 4: Training and validation of the DNN model for shapes of different complexity (m). The Y-axis in each of the plots shows the mean-squared error (MSE) of the model prediction normalized by the variance in the permeability values in each case. As test set we randomly take about 10 % out of the total dataset in each case. The X-axis shows the number of training epochs (iterations).

simple descriptions such as tortuosity and porosity. The volume moment tensor $W_0^{2,0}$ is equivalent to the area moment of inertia tensor and could thus capture effects such as blocking of the flow due to orientation of shapes as is evident from several other studies [25, 26]. However it is not clear how the other MTs may be physically related to the permeability and their applications have remained limited in other applications, as well [8]. In order to show that all the MTs contribute to the prediction of the permeability, we present training results where only one MT is used at a time (6 features) considering the dataset for $m = 2$, in Fig. 5,. For the Y-axis in Fig. 5, the same data and normalization approach were used as in Fig. 4b. Clearly all four MTs are independently able to predict the permeability to great

accuracy and, thus, contribute to the model. The information contained in other well known measures such as *tortuosity* [7] is clearly contained also in the characterisation of the porous microstructure using MTs.

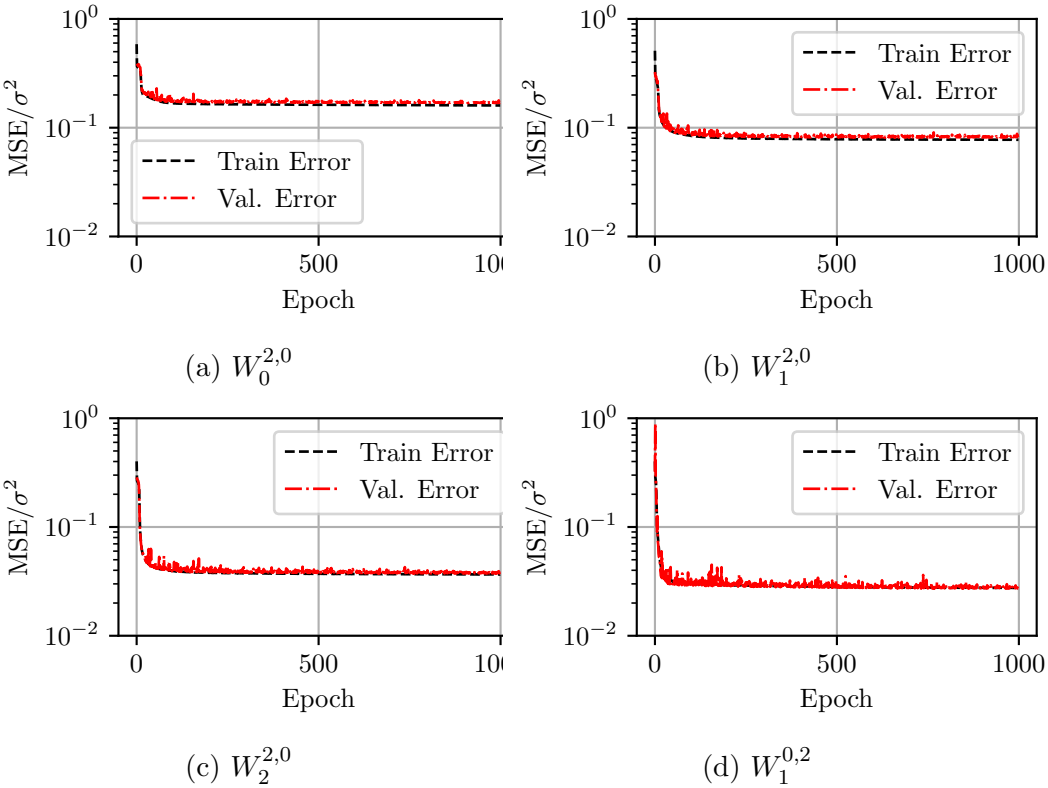


Figure 5: Training and validation of the DNN with different Minkowski tensors used exclusively as features.

It is a practically worthwile question to ask whether a large amount of data may always be required for a successful permeability prediction model. To answer this question we perform the training process with different training data set sizes and juxtapose these results in Fig. 6. Here we plot the minimum error achieved against the size of the training data ($m = 2$). Re-

markably, only a very small number of samples, namely 100, is required to predict the permeability to a mean squared error of 10% w. r. t. variance in the data, for a shape complexity of $m = 2$. This suggests that a similar exercise using three dimensional domains can be achieved with a computationally feasible data size.

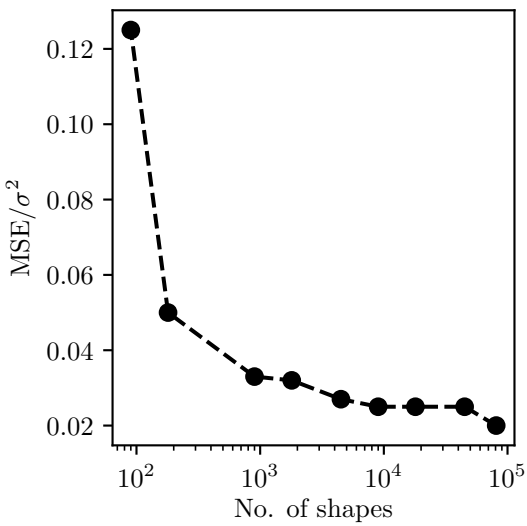


Figure 6: The minimum error in validation data obtained for different training data sizes for the shapes of order $m = 2$. Each point corresponds to a different DNN model obtained with a data size shown on the x-axis. The same training parameters were used. The y-axis shows the mean square error of permeability estimate normalized with total variance in the data for permeability values.

6. Conclusion

We have demonstrated that the answer to the titular question whether Minkowski Tensors of a porous microstructure can be related exclusively to the permeability of the medium is a 'yes.' We have generated a large enough

dataset of complex shapes of constant porosity with different orders of complexity and performed CFD simulations of flows through the REVs containing these non-convex simply connected shapes. We trained a DNN using these simulated data, validated the model, and demonstrated its accuracy for different shape complexity and for different dataset sizes. Reasonably accurate prediction results from even a small number (≈ 100) of data points for training the model. This indicates that computationally more challenging systems can be investigated following the procedure outlined above.

Physically, these unit cells with obstacles may be seen as representing cross sections of uni-directional fibrous porous media and may seem limited in application. However, the motivation to the above exercise has been to identify a simple system where a data experiment relating the MTs to the permeability may be conducted. Our results show that Minkowski tensors are sufficient descriptors of the pore microstructure, in order to determine its permeability. This result leads us to hypothesize that the same may be true for much larger scale systems in three dimensions and involving non-uniform porous microstructure. Further investigations into anisotropic porous media, where the tensorial nature of permeability is important and porous media with higher Euler characteristics (non-simply connected microstructure) are natural extensions to the present work.

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